

Claudio Del Sole

Postdoctoral Researcher in Statistics

Università degli Studi di Milano–Bicocca, Milan, Italy

claudio.delsole@unimib.it

<https://claudiodelsole.github.io/>

Academic Positions

Nov 2024 - present **Postdoctoral researcher**
Università degli Studi di Milano–Bicocca (Milan, Italy)
Department of Economics, Management and Statistics
Supervisor: Federico Camerlenghi.

Education

Sep 2020 - Jan 2025 **PhD in Statistics and Computer Science**
Bocconi University (Milan, Italy)
Advisors: Antonio Lijoi and Igor Prünster.
Thesis title: *Contributions to dependent processes in Bayesian nonparametrics*.
Thesis defense: 29th January 2025.

Sep 2018 - Oct 2020 **Master Degree in Mathematical Engineering**
Politecnico di Torino (Turin, Italy)
Advisors: Paola Siri and Enrico Bibbona.
Thesis title: *Stochastic models of Chemical Reaction Networks: multiscale approximations and convergence*.

Feb 2019 - Oct 2020 **Alta Scuola Politecnica**
Politecnico di Torino (Turin, Italy) & Politecnico di Milano (Milan, Italy)
Honors program with interdisciplinary courses and projects.

Sep 2019 - Feb 2020 **Exchange Program (Erasmus+)**
École Polytechnique Fédérale de Lausanne (Lausanne, Switzerland)
Autumn semester exchange program in Applied Mathematics.

Sep 2015 - Oct 2018 **Bachelor Degree in Mathematics for Engineering**
Politecnico di Torino (Turin, Italy)

Publications

Peer-reviewed journals

- Catalano, M., **Del Sole, C.** (2026). Hierarchical Random Measures without Tables. *Bayesian Analysis* (in press). DOI
- **Del Sole, C.**, Lijoi, A., Prünster, I. (2026). Principled Estimation and Prediction with Competing Risks: a Bayesian Nonparametric Approach. *Journal of the American Statistical Association* (in press). DOI

- Catalano, M., **Del Sole, C.**, Lijoi, A., Prünster, I. (2024). A Unified Approach to Hierarchical Random Measures. *D. Basu Centennial Volume, Sankhya A* 86(S1), 255–287. DOI

Conference proceedings

- Catalano, M., **Del Sole, C.** (2023). A Note on the Dependence Structure of Hierarchical Completely Random Measures. In *Bayesian Statistics, New Generations New Approaches*, Springer Proceedings in Mathematics and Statistics, Vol. 435, 77–85. DOI

PhD Thesis

- **Del Sole, C.** (2025). Contributions to dependent processes in Bayesian nonparametrics. IRIS

Working papers

- Beraha, M., Camerlenghi, F., **Del Sole, C.** (2026+). A Bayesian nonparametric approach to the multi-armed bandit problem for traits allocation models.
- **Del Sole, C.**, Lijoi, A., Prünster, I. (2026+). A principled approach to covariate-dependent random partition models.

Funding

Sep 2024 - Oct 2024	Special financial support , Bocconi University (Milan, Italy).
Sep 2020 - Aug 2024	PhD Scholarship , Bocconi University (Milan, Italy).
Oct 2018 - Oct 2020	Alta Scuola Politecnica , Politecnico di Torino (Turin, Italy). Merit-based tuition waver (2 years).
Sep 2015 - Sep 2018	Percorso Giovani Talenti , Politecnico di Torino (Turin, Italy). Merit-based partial tuition waver (3 years) with additional training activities.

Travel awards

July 2026	4th BNP Networking Workshop (Seoul, South Korea).
June 2025	14th International Conference on Bayesian Nonparametrics (Los Angeles, USA).
July 2024	3rd BNP Networking Workshop (Singapore).

Presentations and Conferences

Invited Presentations

July 2026	4th BNP Networking Workshop (Seoul, South Korea).
June 2025	StaTalk (Milan, Italy).
January 2025	International Conference on Recent Developments in the Techniques of Bayesian Paradigm (Varanasi, India).
July 2024	ISBA World Meeting (Venice, Italy).
June 2024	4th Italian Meeting on Probability and Mathematical Statistics (Rome, Italy).
August 2023	6th International Conference on Econometrics and Statistics (Tokyo, Japan).

Contributed Presentations

June 2025 14th International Conference on Bayesian Nonparametrics (Los Angeles, USA).
July 2024 3rd BNP Networking Workshop (Singapore).
October 2023 Autumn School in Bayesian Statistics at CIRM (Marseille, France).

Poster Presentations

June 2026 ISBA World Meeting (Nagoya, Japan).
September 2025 SISBayes Workshop (Padua, Italy).
May 2025 Early-Career Workshop on Nonparametric Statistics (Rome, Italy).
March 2025 10th Survival Analysis for Junior Researchers Conference (Bonn, Germany).
June 2024 BAYSM Bayesian Young Statisticians Meeting (Venice, Italy).
May 2024 StaTalk (Florence, Italy). *Recipient of the Honourable Mention.*
June 2023 Workshop on Approximation Methods in Bayesian Analysis (Marseille, France).

Conference Participations

May 2026 StaTalk (Turin, Italy).
April 2023 Workshop on Theory for Scalable, Modern Statistical Methods (Milan, Italy).
October 2022 13th International Conference on Bayesian Nonparametrics (Puerto Varas, Chile).
April 2022 1st BNP Networking Workshop (Nicosia, Cyprus).

Residential Research Programs

9 July - 2 August 2024 Invited Participant, *Interpretable Inference via Principled BNP Approaches in Biomedical Research and Beyond*, Institute of Mathematical Sciences (IMS), National University of Singapore (Singapore).

Research Visits

March 2026 Department of Statistics, Duke University (Durham, USA).

Research Projects

2023-2025 Member of the European Union – Next Generation EU, PRIN-PNRR 2023-2025 project (P2022H5WZ9) *Measuring Biodiversity via Bayesian Nonparametrics: Estimation, Clustering and Uncertainty Quantification*. PI: Igor Prünster.

Teaching Experience

Instructor

- *Statistics and Probability* (22h), Bocconi University (2026/2027).
- *Livellamento di Analisi* (in Italian) (15h), Università degli Studi di Milano-Bicocca (2025/2026).
- *Mathematical Methods – Module: Probability* (40h), Politecnico di Torino (2022/2023).

Teaching Assistant

- *Mathematical Statistics*, Bocconi University (from 2021/2022 to 2023/2024).
- *Machine Learning II*, Bocconi University (from 2023/2024 to 2025/2026).
- *Advanced Statistical Methods*, Bocconi University (from 2023/2024 to 2025/2026).
- *Advanced Statistics for Economics and Social Sciences*, Bocconi University (from 2023/2024, to 2025/2026).
- *Stochastic Processes*, Bocconi University (from 2023/2024 to 2025/2026).
- *Mathematical Analysis II*, Bocconi University (from 2021/2022 to 2023/2024).
- *Mathematical Methods in Computer Science*, Bocconi University (2023/2024, 2024/2025).
- *Elements of Real and Complex Analysis*, Bocconi University (2022/2023).
- *Principles of Business Analytics*, Bocconi University (from 2022/2023 to 2025/2026).

Academic Tutor

- *Mathematical Methods for Engineering*, Politecnico di Torino (2018/2019).
- *Probability and Statistics*, Politecnico di Torino (2018/2019).

Teaching Material

- **Del Sole, C.**, Lavenant, H. *Mathematical Analysis - Module II, Lecture Notes*. BAI Series, Volume 4. Second Edition - 2023/2024. ISBN: 978-88-6407-515-0.

Training courses

Autumn 2021 **Bocconi Excellence in Advanced Teaching (BEAT)**, Bocconi University Innovation in Learning and Teaching (BUILT), Bocconi University.

Professional Activities

Editorial Work

- Reviewer for *Bayesian Analysis* (4), *Biometrical Journal* (1).

Coordinator, biweekly internal seminar series, BayesLab, Bocconi University.

Skills and Languages

Computing skills

- Public Github repository and Julia package `CompetingRisks.jl` for the paper *Principled Estimation and Prediction with Competing Risks: a Bayesian Nonparametric Approach*.
- Public Github repository and Julia package `hCRV.jl`, including an R interface, for the paper *Hierarchical Random Measures without Tables*.
- **Programming Languages**: Julia, Python, R, Matlab, C++.

Languages

- Italian: mother tongue.
- English: IELTS certificate, Band 8.5/9.0 (November 2019).

Research Networks and Scientific Societies

Research Networks

2023 - present **Research Affiliate**, Bayesian Learning Laboratory (BayesLab), Bocconi Institute for Data Science and Analytics (BIDSA), Bocconi University.

Society Memberships

- International Society for Bayesian Analysis (ISBA). Sections: Bayesian Nonparametrics, Junior-ISBA.
- Italian Statistical Society (SIS). Sections: y-SIS, SISBayes.